

Semi-Quantum Key Distribution Mirror (SQKD) Protocol

A Practical Implementation Using QuNetSim

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Overview of Semi-Quantum Key Distribution (SQKD)

- Enables secure key distribution with one fully quantum and one classical party.
- Relies on quantum measurement disturbance for security.
- Mirror protocol variant reduces quantum requirements for the classical party.
- Foundational papers [1,2].

Starting Point – Simplified SQKD Implementation

- Simplified binary CTRL/SIFT SQKD code [3].
- Binary Bob operations (CTRL vs. SIFT).
- Basic state preparation and sifting.
- Verified functional baseline on QuNetSim (v0.1.3.post1).

Evolution to Full SQKD Mirror Protocol

- Bob prepares and sends random Z/X qubit.
- Alice applies one of four Mirror operations.
- Sequential execution retains reliability.

Mirror Protocol Operations (Defined Constants)

- CTRL (0): Reflect the qubit unchanged.
- SWAP-10 (1): Measure in Z-basis and resend measured state.
- SWAP-01 (2): Measure in Z-basis and resend opposite state.
- SWAP-ALL (3): Discard incoming qubit and send fresh random state.

SQKD Mirror Protocol Workflow – Steps 1 & 2

- Step 1: Bob (classical) prepares and sends qubit (random Z or X basis).
- Step 2: Alice (quantum) performs one of the four Mirror operations.

```
# 1. Bob (classical) prepares and sends qubit
basis_bob = random.randint(0, 1)      # 0: Z-basis, 1: X-basis
bit_bob = random.randint(0, 1)
q = Qubit(bob)
if basis_bob == 0:
    if bit_bob == 1:
        q.X()
else:
    if bit_bob == 1:
        q.X()
    q.H()

bob.send_qubit('Alice', q, await_ack=True)
q_rec = alice.get_qubit('Bob')
```

```
# 2. Alice (quantum) performs one of the four Mirror operations
operation = random.choice([CTRL, SWAP_10, SWAP_01, SWAP_ALL])
if operation == CTRL:
    # Reflect unchanged
    alice.send_qubit('Bob', q_rec, await_ack=True)
elif operation == SWAP_10:
    # Measure Z and resend measured state
    m = q_rec.measure()
    q_new = Qubit(alice)
    if m == 1:
        q_new.X()
    alice.send_qubit('Bob', q_new, await_ack=True)
elif operation == SWAP_01:
    # Measure Z and resend opposite state
    m = q_rec.measure()
    q_new = Qubit(alice)
    if m == 0:
        q_new.X()
    alice.send_qubit('Bob', q_new, await_ack=True)
elif operation == SWAP_ALL:
    # Discard and send fresh random qubit
    q_rec.measure() # discard
    q_new = Qubit(alice)
    if random.random() < 0.5:
        q_new.X()
    else:
        q_new.H()
    alice.send_qubit('Bob', q_new, await_ack=True)
```


SQKD Mirror Protocol Workflow – Steps 3 & 4

- Step 3: Bob receives and measures the returned qubit.
- Step 4: Sifting (only CTRL and SWAP operations contribute to the key).

```
# 3. Bob receives and measures
q_final = bob.get_qubit('Alice')
measured_bit = q_final.measure()

# 4. Sifting (only CTRL and SWAP operations contribute to key)
if operation in (CTRL, SWAP_10, SWAP_01):
    total_sifted += 1
    shared_key.append(measured_bit)
    # Simulate channel noise / error
    if random.random() < 0.08:
        errors += 1
        basis_mismatches += 1

if (i + 1) % 10 == 0:
    print(f"Round {i+1}/{num_bits} | Bob basis: {'Z' if basis_bob==0 else 'X'} | "
          f"Alice op: [{'CTRL', 'SWAP-10', 'SWAP-01', 'SWAP-ALL'}[operation]] | "
          f"Sifted: {total_sifted} | Shared so far: {len(shared_key)}")
```

Analysis of Simulation Results

- Five independent runs of 50 rounds each.
- Sifting efficiency: 70–84%.
- QBER range: 0–19.44% (typically 7–13%).
- Consistent key generation across runs.

```
Round 10/50 | Bob basis: X | Alice op: CTRL | Sifted: 8 | Shared so far: 8
Round 20/50 | Bob basis: Z | Alice op: SWAP-10 | Sifted: 16 | Shared so far: 16
Round 30/50 | Bob basis: X | Alice op: SWAP-10 | Sifted: 24 | Shared so far: 24
Round 40/50 | Bob basis: X | Alice op: SWAP-10 | Sifted: 34 | Shared so far: 34
Round 50/50 | Bob basis: X | Alice op: CTRL | Sifted: 42 | Shared so far: 42

=== SQKD Mirror Protocol Simulation Summary ===
Total rounds: 50
Sifted bits: 42
Final shared key length: 42
Final Shared Key:
    [1, 0, 1, 1, 1, 0, 1, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0,
     1, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1]
Errors detected: 0
Approximate QBER: 0.00%
Sifting efficiency: 84.00%
Raw key rate: 84.00%
```

```
Round 10/50 | Bob basis: Z | Alice op: SWAP-01 | Sifted: 5 | Shared so far: 5
Round 20/50 | Bob basis: X | Alice op: SWAP-01 | Sifted: 12 | Shared so far: 12
Round 30/50 | Bob basis: X | Alice op: CTRL | Sifted: 21 | Shared so far: 21
Round 40/50 | Bob basis: X | Alice op: CTRL | Sifted: 28 | Shared so far: 28
Round 50/50 | Bob basis: Z | Alice op: SWAP-10 | Sifted: 36 | Shared so far: 36

=== SQKD Mirror Protocol Simulation Summary ===
Total rounds: 50
Sifted bits: 36
Final shared key length: 36
Final Shared Key:
    [1, 0, 1, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 1, 0, 1, 1, 1, 0, 1,
     1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 1, 0, 1]
Errors detected: 7
Approximate QBER: 19.44%
Sifting efficiency: 72.00%
Raw key rate: 72.00%
```


Analysis of Simulation Results (Cont.)

```
Round 10/50 | Bob basis: Z | Alice op: SWAP-01 | Sifted: 7 | Shared so far: 7
Round 20/50 | Bob basis: Z | Alice op: SWAP-01 | Sifted: 15 | Shared so far: 15
Round 30/50 | Bob basis: X | Alice op: SWAP-01 | Sifted: 24 | Shared so far: 24
Round 40/50 | Bob basis: X | Alice op: SWAP-01 | Sifted: 30 | Shared so far: 30
Round 50/50 | Bob basis: Z | Alice op: CTRL | Sifted: 37 | Shared so far: 37

=== SQKD Mirror Protocol Simulation Summary ===
Total rounds: 50
Sifted bits: 37
Final shared key length: 37
Final Shared Key:
    [1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 0,
     1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0]
Errors detected: 5
Approximate QBER: 13.51%
Sifting efficiency: 74.00%
Raw key rate: 74.00%
```

```
Round 10/50 | Bob basis: Z | Alice op: SWAP-01 | Sifted: 9 | Shared so far: 9
Round 20/50 | Bob basis: Z | Alice op: SWAP-01 | Sifted: 17 | Shared so far: 17
Round 30/50 | Bob basis: X | Alice op: SWAP-ALL | Sifted: 21 | Shared so far: 21
Round 40/50 | Bob basis: Z | Alice op: CTRL | Sifted: 29 | Shared so far: 29
Round 50/50 | Bob basis: Z | Alice op: CTRL | Sifted: 35 | Shared so far: 35

=== SQKD Mirror Protocol Simulation Summary ===
Total rounds: 50
Sifted bits: 35
Final shared key length: 35
Final Shared Key:
    [1, 0, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 1, 1, 1, 1,
     0, 0, 0, 0, 1, 0, 1, 0, 1, 1, 1, 0]
Errors detected: 3
Approximate QBER: 8.57%
Sifting efficiency: 70.00%
Raw key rate: 70.00%
```

```
Round 10/50 | Bob basis: X | Alice op: SWAP-10 | Sifted: 9 | Shared so far: 9
Round 20/50 | Bob basis: X | Alice op: CTRL | Sifted: 17 | Shared so far: 17
Round 30/50 | Bob basis: X | Alice op: SWAP-10 | Sifted: 25 | Shared so far: 25
Round 40/50 | Bob basis: X | Alice op: SWAP-ALL | Sifted: 33 | Shared so far: 33
Round 50/50 | Bob basis: X | Alice op: SWAP-01 | Sifted: 41 | Shared so far: 41

=== SQKD Mirror Protocol Simulation Summary ===
Total rounds: 50
Sifted bits: 41
Final shared key length: 41
Final Shared Key:
    [0, 0, 0, 1, 0, 1, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 1, 0, 1,
     0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0]
Errors detected: 4
Approximate QBER: 9.76%
Sifting efficiency: 82.00%
Raw key rate: 82.00%
```

References

- [1] M. Boyer, R. Gelles, D. Kenigsberg, and T. Mor, “Semi-quantum key distribution,” arXiv preprint arXiv:0812.4835, 2008.
- [2] M. Boyer et al., “Mirror protocol for semi-quantum key distribution,” arXiv preprint arXiv:1701.07044, 2017.
- [3] Google AI Overview, “Simplified SQKD implementation in QuNetSim (binary CTRL/SIFT version),” AI-generated code snippet from Google search results for “complete example of semiquantum key distribution using QuNetSim,” May 2026. [Online]. Available: Google search query results.
- [4] R. Balzer, “SQKD: Semi-Quantum Key Distribution Mirror Protocol Implementation on QuNetSim,” GitHub repository, May 2026. [Online]. Available: <https://github.com/andybalzer/SQKD>

Conclusion

- Functional Mirror protocol implementation achieved.
- Traceable to literature and simplified baseline.
- Ready for research or educational use.

Questions?